MSC 201 **59359**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It uses the sample size. B. It retains mass units. C. It assumes independent variation. D. It predicts a nonnegative SD. 10 pts
2. For a normal model, where is the 10th percentile?	A. Below the mean. B. At the largest possible value. C. At the mean. D. Above the mean. 10 pts
3. A sample's mean file size is 18 KB. The study aims to estimate the mean across all files on a server. What does 18 KB represent?	A. The exact population mean regardless of selection. B. The size of every file on the server. C. An estimate based on the sampled files. D. A guaranteed maximum file size. 10 pts
4. How does no arrivals in the next 5 minutes relate to waiting more than 5 minutes for the next arrival?	A. They describe the same event, ignoring a zero-probability boundary. B. The first requires exactly one arrival. C. They are independent unrelated events. D. They are complements. 10 pts
5. A supervisor predicts the expected total number of jobs across many sessions. What does that prediction describe?	A. The exact total in the next batch. B. The average total across repeated batches of sessions. C. The most jobs any batch could contain. D. The most likely total in every possible model. 10 pts
6. A sample has 2 successes and 98 failures. Does a large total alone justify the ordinary normal-approximation proportion interval?	A. Yes; n=100 guarantees every condition. B. Yes; failures are numerous. C. No; the observed success count is too sparse. D. No; proportions cannot be estimated at all. 10 pts

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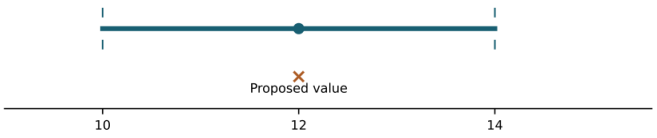
MSC 201 **59359**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Two successes in four trials can occur in several orders. Why must all those orders be considered?	A. The count event includes every order with two successes. B. The expected count is an order. C. Only the first successful order can occur. D. Different orders are the same elementary outcome. 10 pts
8. A bell-shaped distribution has mean 20 and SD 2. Approximately which interval contains 95%?	A. 20 to 24. B. 0 to 20. C. 18 to 22. D. 16 to 24. 10 pts
9. P(A)=0.2 but P(A AND B)=0.3. Why is this impossible?	A. The overlap must always equal zero. B. The overlap cannot exceed the probability of A. C. A probability cannot be below 0.5. D. B must have probability 1. 10 pts
10. A mean interval is 10 to 14. How should a proposed mean of 12 be described? 	A. Impossible. B. The only compatible value. C. Proved to be the exact population mean. D. Compatible with the interval, without being proved exact. 10 pts

59359

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3.

C.

10 pts

4. A.

10 pts

5.

B.

10 pts

6.

C.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

D.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **48688**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Data are 0, 8, 9, 9, 10. Which center is lower?	A. The median. B. They are equal. C. The maximum. D. The mean. 10 pts
2. The slides recommend plus-four when confidence is at least 90% and n>=10. Does n=24 at 85% confidence meet that recommendation?	A. Yes; n alone is sufficient. B. No; the confidence level is below the stated threshold. C. No; n must exceed 100. D. Yes; plus-four has no stated conditions. 10 pts
3. 20 students use both devices; 40 use tablets and 80 use laptops. Which conditional probability is larger?	A. Laptop use given tablet use. B. They must be equal. C. Tablet use given laptop use. D. Neither can use the overlap count. 10 pts
4. A symmetric 90% confidence construction leaves what fraction in each tail?	A. 0.90. B. 0.45. C. 0.05. D. 0.10. 10 pts
5. A payout variable is measured in dollars. What units belong to its variance?	A. Dollars. B. Percent. C. No units. D. Dollars squared. 10 pts
6. A sample grows from 50 to 51 observations. Must its mean move closer to the population mean?	A. Yes; every increase improves the estimate. B. No; larger samples always worsen estimates. C. Yes; the law of large numbers guarantees monotone improvement. D. No; individual additions can move it either way. 10 pts
7. A roster repeats a ten-person pattern, with every tenth entry a team leader. Why could taking every tenth entry be problematic?	A. Systematic selection always includes everyone. B. A larger interval always makes selection random. C. The sampling interval can align with the repeating pattern. D. Team leaders cannot answer surveys. 10 pts

Test continues on the other side of the sheet.

MSC 201 **48688**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A continuous model gives F(6)=0.70. What does that say?	A. 70% of durations equal exactly 6. B. The mean is 6. C. The density height at 6 is 0.70. D. 70% of modeled durations are at most 6. 10 pts
9. P(X>50)=0.50 and P(X>60)=0.10. Given X>50, what is P(X>60)?	A. 0.20. B. 0.10. C. 0.60. D. 0.40. 10 pts
10. A sample is drawn without replacement from a box containing marked and unmarked cards. When must the sample contain a marked card?	A. Only if every card in the box is marked. B. Whenever the sample size is less than the number of unmarked cards. C. Whenever one marked card is present in the box. D. When the sample size exceeds the number of unmarked cards. 10 pts

48688

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3. A.

10 pts

4.

C.

10 pts

5.

D.

10 pts

6.

D.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

D.

10 pts

9. A.

10 pts

10.

D.

10 pts

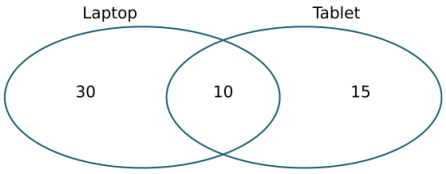
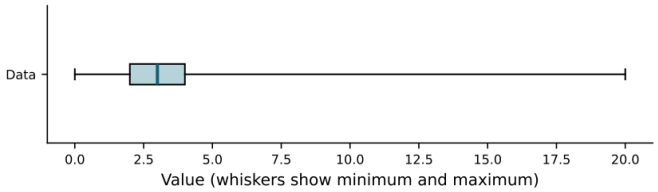
MSC 201 **39587**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For known population SD and fixed confidence, n quadruples. What happens to the margin of error?	A. It doubles. B. It stays unchanged. C. It quarters. D. It halves. 10 pts
2. An exact continuous duration has P(X=4)=0. Can a timer report 4 seconds?	A. No; 4 seconds can never be recorded. B. Yes; rounding collects a positive-width interval of exact values. C. Only if the density equals zero everywhere. D. Only if the model becomes discrete before rounding. 10 pts
3. 40 students use laptops, 25 use tablets, and 10 use both. Why subtract 10 when counting either device? 	A. Both-device users must be excluded entirely. B. The overlap does not use either device. C. Adding the first two counts includes the overlap twice. D. Subtraction is required because the sample is random. 10 pts
4. A basic box plot reports minimum 0, Q1=2, median 3, Q3=4, maximum 20. Which comparison is supported? 	A. The segment from Q3 to the maximum spans a wider numerical range than the segment from the minimum to Q1. B. The plot establishes the exact mean. C. The upper segment must contain three times as many observations. D. The lower segment contains no observations. 10 pts
5. A teacher draws cards without replacement until the first marked card. Why is the draw number generally neither binomial nor geometric?	A. The trial count is not fixed, and the chance can change between draws. B. Stopping at the first success always makes the count binomial. C. The draw number is continuous. D. A success has to be a desirable outcome. 10 pts

Test continues on the other side of the sheet.

MSC 201 **39587**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

6. Why represent a single binomial count by an interval when using a normal approximation?	A. An interval has area under a continuous curve, while a single point has zero area. B. The normal distribution has positive probability at every exact point. C. The count can take fractional values. D. A count always equals its expected value. 10 pts
7. A distribution has equal peaks near 2 and 8 and is symmetric around 5. What follows?	A. Symmetry does not require a single peak. B. It must be right-skewed. C. A mean cannot exist. D. Its mode must be uniquely 5. 10 pts
8. Two normal curves share SD but have different means. What is equal?	A. Their centers. B. Their probabilities below every fixed cutoff. C. Their widths and shapes. D. Their 90th-percentile values. 10 pts
9. Three reported percentages are 33.3%, 33.3%, and 33.3%. What most reasonably explains the 99.9% total?	A. The data must be fabricated. B. Missing exactly one participant. C. Overlapping categories are certain. D. Rounding. 10 pts
10. A value equals its distribution's mean. What is its z-score?	A. The mean. B. 1. C. The SD. D. 0. 10 pts

39587

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4. A.

10 pts

5. A.

10 pts

TEST PAGE 2

6. A.

10 pts

7. A.

10 pts

8.

C.

10 pts

9.

D.

10 pts

10.

D.

10 pts

MSC 201 **62859**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Mean is 30 and SD is 4. Which value has z=1.5?	A. 26. B. 45. C. 31.5. D. 36. 10 pts
2. A box plot has median 6. Can it reveal exactly how many observations equal 5?	A. Yes; none equal 5. B. Yes; half equal 5. C. Yes; exactly one quarter equal 5. D. No; the summary does not retain those exact counts. 10 pts
3. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It retains mass units. B. It assumes independent variation. C. It uses the sample size. D. It predicts a nonnegative SD. 10 pts
4. An exponential density extends far right with diminishing height. What does that say about very long waits?	A. They must occur as often as short waits. B. They are possible but less common than short waits. C. Every long wait has probability 1. D. They are impossible once the mean is exceeded. 10 pts
5. Everyone in a few selected clubs is surveyed. What challenges treating all responses as independent?	A. The sample proportion cannot be computed. B. Club members cannot have binary responses. C. Randomly selecting clubs guarantees individual independence. D. Members of the same club may have correlated responses. 10 pts
6. Campus A has 30 users among 60 students; campus B has 40 among 100. Which has the higher use rate?	A. Campus B. B. They have equal rates. C. Counts alone prove campus B has the higher rate. D. Campus A. 10 pts

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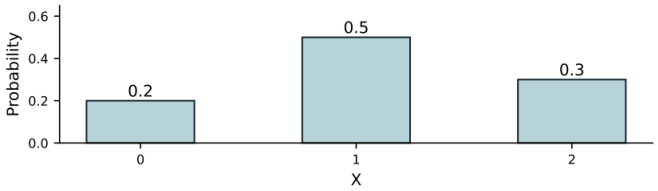
MSC 201 **62859**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 256. B. 15. C. 17. D. 16. 10 pts
8. A researcher stops after a fixed number of draws and counts marked tokens. Which detail best distinguishes the binomial and hypergeometric settings?	A. Whether the recorded result is a whole number. B. Whether marked is a desirable outcome. C. Whether a fixed number of draws is used. D. Whether each token is returned and the pool remixed before the next draw. 10 pts
9. The sample is an unordered set of cables. Why count combinations rather than different orders of the same set?	A. Every sample set has only one cable. B. Combinations assume replacement. C. Reordering the same selected cables does not create a new sample set. D. Selection order determines whether a cable is faulty. 10 pts
10. A discrete probability table has a row at the cutoff. How do the events below the cutoff and at or below the cutoff differ? 	A. Both exclude the row at the cutoff. B. Only the second includes the row at the cutoff. C. They include opposite sides of the cutoff. D. Only the first includes the row at the cutoff. 10 pts

62859

TEST PAGE 1

1.

D.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

D.

10 pts

9.

C.

10 pts

10.

B.

10 pts

MSC 201 **47741**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Daily print totals are shown in a histogram instead of a time-series graph. What information is lost?	A. The existence of high totals. B. Which total occurred on which date. C. The approximate distribution of totals. D. The number of daily observations. 10 pts
2. The plus-four estimate adds two successes and two failures. With 8 successes out of 40, what adjusted fraction is used?	A. 8/44. B. 10/44. C. 10/40. D. 12/44. 10 pts
3. Why is adding signed deviations from the sample mean unsuitable as a spread measure?	A. All deviations are positive. B. The mean is always the largest value. C. It necessarily counts each value twice. D. Positive and negative deviations cancel. 10 pts
4. A researcher stops after a fixed number of draws and counts marked tokens. Which detail best distinguishes the binomial and hypergeometric settings?	A. Whether the recorded result is a whole number. B. Whether marked is a desirable outcome. C. Whether each token is returned and the pool remixed before the next draw. D. Whether a fixed number of draws is used. 10 pts
5. The course uses N(mean, SD). A technician expects readings from N(40,6) to cluster near 6. What is the best correction?	A. They cluster near 6; 40 is the sample size. B. The readings cluster near 40; 6 describes their spread. C. Every reading must equal 40. D. The model allows only the two values 6 and 40. 10 pts
6. Mean is 100, SD is 10, and the supplied 90th-percentile z is 1.28. What is the corresponding value?	A. 87.2. B. 112.8. C. 12.8. D. 101.28. 10 pts
7. Four jobs have a mean of 5 pages. What is the total number of pages?	A. 1.25. B. 9. C. 20. D. 5. 10 pts

Test continues on the other side of the sheet.

MSC 201 **47741**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

8. To find the probability of A followed by B when the stages may be dependent, which expression fits?	A. P(A) times P(B) without checking independence. B. P(A) minus P(B given A). C. P(A) times P(B given A). D. P(A) plus P(B). 10 pts
9. A supplied normal distribution of sample means gives cumulative probabilities 0.15 at 8 and 0.85 at 12. What is the probability between 8 and 12?	A. 0.70. B. 0.15. C. 1.00. D. 0.85. 10 pts
10. Both endpoints of a uniform model increase by 4. What happens to its mean and SD?	A. Both rise by 4. B. Both stay unchanged. C. SD rises by 4; mean stays unchanged. D. Mean rises by 4; SD stays unchanged. 10 pts

47741

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6.

B.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

C.

10 pts

9. A.

10 pts

10.

D.

10 pts

MSC 201 **48040**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A random sample of 80 enrolled students reports 30% support for later hours. Which conclusion is justified?	A. Exactly 24 students in the entire college support later hours. B. Exactly 30% of all students support later hours. C. 30% describes the sample and estimates population support. D. The population percentage must exceed 30%. 10 pts
2. A recorded file size is far above the rest. What should happen before deleting it?	A. Replace it with zero automatically. B. Check whether it is an error or a genuine unusual file. C. Keep it only if the mean equals the median. D. Delete it because all outliers are mistakes. 10 pts
3. One hundred valid 95% intervals are constructed in repeated sampling. Must exactly 95 cover the mean?	A. No; the coverage count itself can vary. B. Yes; any other count proves bias. C. No; all 100 must cover it. D. Yes; the confidence level fixes that count. 10 pts
4. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. Allowing a long wait. B. Nonnegative waiting times. C. Recording waits in minutes. D. A constant rate across the whole period. 10 pts
5. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. They must have equal tails. B. The model with the larger SD. C. The one with more recorded observations regardless of SD. D. The model with the smaller SD. 10 pts
6. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 72. B. 6. C. 2. D. 12. 10 pts

Test continues on the other side of the sheet.

MSC 201 **48040**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. The numbers of pages in six jobs are 1, 2, 2, 4, 4, 5. Which statement about the mode is correct?	A. Only 4 is a mode. B. Both 2 and 4 are modes. C. The mode is 3, halfway between 2 and 4. D. Only 2 is a mode. 10 pts
8. A small prize is common and a large prize is rare. Why is the midpoint between the prizes generally a poor estimate of the expected payout?	A. It gives the two prizes equal weight despite their different chances. B. The expected payout must equal the larger prize. C. Only the larger prize determines the mean. D. The small prize does not count toward the average. 10 pts
9. For a standard normal variable, compare $P(Z<-1.4)$ and $P(Z>1.4)$.	A. They are equal. B. The left tail is larger. C. Both equal 0.5. D. The right tail is larger. 10 pts
10. Two mutually exclusive events each have positive probability. Can they be independent?	A. Yes; both have positive probabilities. B. Yes; disjoint always means independent. C. No; one occurring makes the other impossible. D. Only if their probabilities add to 1. 10 pts

48040

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3. A.

10 pts

4.

D.

10 pts

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B.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

B.

10 pts

8. A.

10 pts

9. A.

10 pts

10.

C.

10 pts

MSC 201 **38655**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A count variable has expected value 0.4. What does that mean?	A. Exactly 40% of sessions must contain one event. B. The model is invalid. C. Its long-run average can be 0.4 despite integer observations. D. A single session must contain 0.4 events. 10 pts
2. On a die, A is even and B is greater than 3. Which results satisfy at least one condition?	A. 1, 3 only. B. 2, 4, 5, 6. C. 4, 6 only. D. All six results. 10 pts
3. Remove 100 from 1, 2, 3, 4, 100. What happens to the mean and median?	A. Both decrease, with a much larger decrease in the mean. B. The mean increases and median decreases. C. The median increases and mean stays unchanged. D. Both stay unchanged. 10 pts
4. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The true sample mean can actually be below every observation. B. The observations must include negative waits. C. It is approximation error outside the true support. D. The normal model is exact at every tail. 10 pts
5. A narrow interval uses only the fastest successful jobs to estimate the mean for all jobs. What is its main limitation?	A. Its narrow width guarantees representativeness. B. Adding decimal places removes selection bias. C. The mean cannot describe time. D. Biased selection can make the interval misleading despite narrow width. 10 pts
6. Two successes in four trials can occur in several orders. Why must all those orders be considered?	A. Different orders are the same elementary outcome. B. Only the first successful order can occur. C. The count event includes every order with two successes. D. The expected count is an order. 10 pts

Test continues on the other side of the sheet.

MSC 201 **38655**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

7. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. Nonnegative waiting times. B. Recording waits in minutes. C. Allowing a long wait. D. A constant rate across the whole period. 10 pts
8. A symmetric interval is 8 to 12. What is the margin of error?	A. 2. B. 4. C. 12. D. 10. 10 pts
9. Campus A has 30 users among 60 students; campus B has 40 among 100. Which has the higher use rate?	A. Counts alone prove campus B has the higher rate. B. They have equal rates. C. Campus B. D. Campus A. 10 pts
10. A recorded value has z=-1.5. What does that mean?	A. It is 1.5 original units below zero. B. It has probability -1.5. C. It is 1.5 SD below the mean. D. It is above 98.5% of values automatically. 10 pts

38655

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3. A.

10 pts

4.

C.

10 pts

5.

D.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

D.

10 pts

8. A.

10 pts

9.

D.

10 pts

10.

C.

10 pts

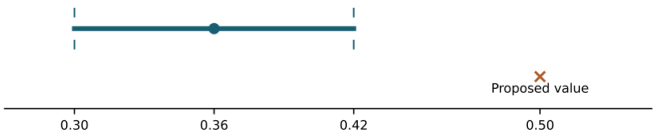
MSC 201 **62608**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A spinner has three colors occupying half, one quarter, and one quarter of its area. Why is assigning each color probability 1/3 inappropriate?	A. A spinner cannot have three outcomes. B. The color areas are unequal. C. Every color must have probability 1/2. D. Probabilities must be whole numbers. 10 pts
2. What equation describes a continuous model's 90th percentile k?	A. The mean equals 0.90k. B. P(X=k)=0.90. C. P(X>k)=0.90. D. P(X<=k)=0.90. 10 pts
3. In 2, 3, 3, 4, 18, the 18 is replaced by 180. What happens?	A. The mean increases; the median stays 3. B. The median increases more than the mean. C. The mean decreases. D. Both stay unchanged. 10 pts
4. A population proportion interval is 0.30 to 0.42. How should a proposed proportion 0.50 be described? 	A. Outside the interval and incompatible with this interval estimate. B. Compatible because 0.50 is a valid probability. C. Proved impossible under every future sample. D. The required sample estimate. 10 pts
5. A value equals its distribution's mean. What is its z-score?	A. 0. B. The SD. C. 1. D. The mean. 10 pts
6. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. The expected count is an integer. B. Every trial must succeed. C. Success probability exceeds 0.5. D. Fixed trial count, two outcomes, independence, and constant success chance. 10 pts

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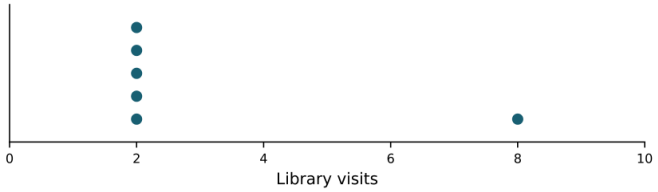
MSC 201 **62608**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. F(3)=0.20 and F(7)=0.80 for a continuous variable. What is P(3<X<7)?	A. 0.80. B. 0.60. C. 1.00. D. 0.20. 10 pts
8. A dot plot has five dots at 2 visits and one dot at 8 visits. What does the stack at 2 mean? 	A. One student reported 10 visits. B. Five students reported 2 visits each. C. Five students reported 8 visits each. D. The average is necessarily 2. 10 pts
9. For independent observations with finite variance, compare repeated means at n=1 and n=100. What is generally expected?	A. The n=1 histogram must be normal. B. Means at n=100 are less spread and often more nearly normal. C. Every size-100 mean is identical. D. The population itself becomes less spread. 10 pts
10. A sample has variance 16 seconds squared. What is its standard deviation?	A. 4 seconds. B. 4 seconds squared. C. 8 seconds. D. 16 seconds. 10 pts

62608

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3. A.

10 pts

4. A.

10 pts

5. A.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

B.

10 pts

9.

B.

10 pts

10. A.

10 pts

MSC 201 **70294**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. The ordered data are 2, 4, 4, 7, 20. What is the median?	A. 20. B. 4. C. 6. D. 7. 10 pts
2. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 256. B. 17. C. 16. D. 15. 10 pts
3. P(A)=0.4, P(B)=0.5, and their overlap is 0.2. What is P(A OR B)?	A. 0.7. B. 0.9. C. 1.1. D. 0.2. 10 pts
4. Notebook colors are blue, blue, black, red. Which statement meaningfully summarizes a typical category?	A. The mean color is 2. B. The median is 2.5 colors. C. The standard deviation of color names is 1. D. Blue is the most common color. 10 pts
5. The SD of a normal model increases while the mean stays fixed. What happens to its peak?	A. It becomes higher and narrower. B. The total area rises above 1. C. The center shifts right. D. It becomes lower as the curve spreads out. 10 pts
6. A student treats z=2 as a probability of 2. What is wrong?	A. A z-score can never exceed 1. B. A standardized distance is not an area probability. C. A z-score must be a percentage. D. A normal model cannot have positive z-scores. 10 pts
7. For a normal approximation to the probability of at most a given count, which intervals belong in the shaded area?	A. The cutoff count's interval and all intervals below it. B. Only the cutoff count's interval. C. Only the intervals above the cutoff count's interval. D. Only the intervals below the cutoff count's interval. 10 pts

Test continues on the other side of the sheet.

MSC 201 **70294**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Plan A samples students from every building. Plan B randomly chooses two buildings and surveys everyone there. What distinguishes Plan B?	A. It necessarily includes every building. B. It selects some students from every group. C. It studies all students in selected groups. D. It guarantees all possible student samples are equally likely. 10 pts
9. The wait until the next request is modeled as exponential with rate 4 per hour. What is its mean in minutes?	A. 240 minutes. B. 4 minutes. C. 60 minutes. D. 15 minutes. 10 pts
10. X counts broken cables in independent checks. Can broken be called success in a binomial model?	A. Only if no cables are broken. B. No; success must be desirable. C. Yes; success means the counted outcome. D. Only if more than half are broken. 10 pts

70294

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

B.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

C.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **60361**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A supplied positive critical z leaves 0.025 to its right. What cumulative area lies to its left?	A. 0.5. B. 0.975. C. 0.95. D. 0.025. 10 pts
2. A value equals its distribution's mean. What is its z-score?	A. 1. B. 0. C. The mean. D. The SD. 10 pts
3. Randomly assigned groups have slightly different initial mean scores. What follows?	A. The study must have used a census. B. Chance imbalance can occur despite random assignment. C. The assignment was necessarily dishonest. D. Random assignment requires exact matching of every score. 10 pts
4. A one-sample t interval uses $n=16$. What degrees of freedom are used?	A. 256. B. 17. C. 16. D. 15. 10 pts
5. Four cables are sampled uniformly without replacement from 12, of which 5 are faulty. Which model directly fits the faulty count?	A. Geometric waiting count. B. Hypergeometric. C. Continuous uniform. D. Normal individual measurement. 10 pts
6. An exponential model has mean wait 4 minutes. What is its SD?	A. 16 minutes. B. 0 minutes. C. 2 minutes. D. 4 minutes. 10 pts
7. SE is 4 at $n=25$. What is SE at $n=100$, with the same population SD and independence?	A. 8. B. 1. C. 2. D. 4. 10 pts

Test continues on the other side of the sheet.

MSC 201 **60361**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Score A is 10 above its mean with SD 5; score B is 12 above its mean with SD 12. Which is relatively farther above its group mean?	A. A. B. B. C. The larger raw score difference always wins. D. They are equally far. 10 pts
9. An independent fair coin has just produced five tails. What is the chance of heads next?	A. 1/2. B. Less than 1/2 because tails is favored. C. Greater than 1/2 because heads is due. D. 1. 10 pts
10. In 2, 3, 3, 4, 18, the 18 is replaced by 180. What happens?	A. The mean decreases. B. The median increases more than the mean. C. Both stay unchanged. D. The mean increases; the median stays 3. 10 pts

60361

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6.

D.

10 pts

7.

C.

10 pts

TEST PAGE 2

8. A.

10 pts

9. A.

10 pts

10.

D.

10 pts

MSC 201 **65708**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A game's expected prize is 4 dollars. Which entry fee makes expected net gain zero?	A. 0 dollars. B. 8 dollars. C. 4 dollars. D. The maximum possible prize regardless of its probability. 10 pts
2. Compare one exponential duration with the average of many independent durations. What remains the same?	A. The maximum possible duration becomes finite for the average. B. The mean. C. The SD. D. The entire density shape. 10 pts
3. Four cables are sampled uniformly without replacement from 12, of which 5 are faulty. Which model directly fits the faulty count?	A. Continuous uniform. B. Hypergeometric. C. Geometric waiting count. D. Normal individual measurement. 10 pts
4. A systematic sample starts at ID 3 and takes every tenth ID. Which IDs appear first?	A. 3, 13, 23. B. 10, 20, 30. C. 3, 10, 20. D. 3, 6, 9. 10 pts
5. 40 of 100 sampled jobs succeed. What is the sample success proportion?	A. 2.5. B. 0.60. C. 0.40. D. 40. 10 pts
6. For 2, 3, 3, 4, 18, how does the mean compare with the median?	A. They are equal. B. The mean is smaller. C. The mean is larger. D. No comparison is possible without a calculator. 10 pts
7. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It predicts a nonnegative SD. B. It uses the sample size. C. It retains mass units. D. It assumes independent variation. 10 pts

Test continues on the other side of the sheet.

MSC 201 **65708**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Tokens are replaced and mixed after each draw. Compared with leaving them out, what does replacement preserve?	A. A decreasing total number of tokens. B. The same token being selected every time. C. A guarantee that colors alternate. D. The original color probabilities on each independent draw. 10 pts
9. P(Z>1.2)=0.1151. By symmetry, what is P(Z<-1.2)?	A. -0.1151. B. 0.5. C. 0.8849. D. 0.1151. 10 pts
10. A sample-size formula returns 62.2 observations. Which planned integer size meets that minimum?	A. 61. B. 62. C. 63. D. 60. 10 pts

65708

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4. A.

10 pts

5.

C.

10 pts

6.

C.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **78928**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A U(0,12) wait has already lasted 11 minutes. What is the largest possible additional wait under this model?	A. 11 minutes. B. 12 minutes. C. 1 minute. D. There is no upper bound. 10 pts
2. Two binomial settings have the same fixed number of trials. One has a greater chance of success on each trial. What changes?	A. Both settings have the same expected number of successes. B. Every run in that setting yields more successes. C. The maximum possible number of successes is greater in that setting. D. The expected number of successes is greater in that setting. 10 pts
3. A file is 60 KB; its group has mean 50 KB and SD 5 KB. How far above the mean is it?	A. 0.5 standard deviations. B. 2 standard deviations. C. 10 standard deviations. D. 12 standard deviations. 10 pts
4. A strongly skewed variable is converted to z-scores. Does it become normal?	A. Yes; subtracting the mean removes the tail. B. No; z-scores cannot be negative. C. Yes; every z-score distribution is standard normal. D. No; standardization shifts and rescales without removing skew. 10 pts
5. A report gives sample SD=6 minutes and SE=1 minute. Which describes variation among individual observed times?	A. 6 minutes. B. 1 minute. C. The confidence level. D. 7 minutes. 10 pts
6. Why report a confidence interval instead of only a sample mean?	A. It lists every individual observation. B. It guarantees the true mean is the midpoint. C. It expresses uncertainty in estimating the population mean. D. It removes sampling bias. 10 pts

Test continues on the other side of the sheet.

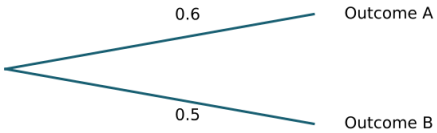
MSC 201 **78928**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Students rate a service while its manager watches. Compared with anonymous responses, what might happen?	A. Sample size automatically doubles. B. Every rating must become negative. C. Students may avoid negative ratings. D. The manager's presence guarantees accuracy. 10 pts
8. A game charges a fixed entry fee and sometimes awards a prize. What distinguishes net gain from the prize amount?	A. Net gain accounts for the fee paid to play. B. Net gain ignores outcomes with no prize. C. The fee changes the chance of each prize. D. Net gain equals the largest possible prize. 10 pts
9. For independent items, the SD of a sum is sqrt(n) times individual SD. Why is n times SD generally wrong?	A. The total is always fixed. B. Means cannot be added. C. All individual deviations have the same sign. D. Independent variances add; SD is then a square root. 10 pts
10. At a tree node, two exhaustive outgoing branch probabilities are 0.6 and 0.5. What is wrong? 	A. Branches cannot represent conditional probabilities. B. All tree branches must be 0.5. C. They should both be zero. D. They total more than 1. 10 pts

78928

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5. A.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

C.

10 pts

8. A.

10 pts

9.

D.

10 pts

10.

D.

10 pts

MSC 201 **65093**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Two servers share one power supply. Why might independent failure events be a poor model?	A. Failure probabilities must sum to 1. B. Different server names prove independence. C. Shared equipment guarantees no failures. D. A power failure can affect both servers together. 10 pts
2. P(Z>1.2)=0.1151. By symmetry, what is P(Z<-1.2)?	A. -0.1151. B. 0.8849. C. 0.1151. D. 0.5. 10 pts
3. The CLT supports an approximately normal distribution of large-sample means. What does it not claim?	A. Means can have approximately normal sampling distributions. B. Individual observations become normally distributed. C. Sampling distributions concern statistics. D. Larger independent samples can have smaller SE. 10 pts
4. A sample has variance 16 seconds squared. What is its standard deviation?	A. 4 seconds. B. 16 seconds. C. 4 seconds squared. D. 8 seconds. 10 pts
5. A selected student's name is removed before the next random draw. What changes?	A. Selection has become a census. B. That student cannot be selected again. C. Every remaining student becomes impossible to select. D. The same name must be selected next. 10 pts
6. A proposed normal model gives substantial probability to negative durations. What is the concern?	A. Every duration must be negative. B. Normal distributions cannot model any positive variable. C. It assigns appreciable probability to impossible values. D. Its mean must be zero. 10 pts

Test continues on the other side of the sheet.

MSC 201 **65093**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A researcher stops after a fixed number of draws and counts marked tokens. Which detail best distinguishes the binomial and hypergeometric settings?	A. Whether a fixed number of draws is used. B. Whether the recorded result is a whole number. C. Whether each token is returned and the pool remixed before the next draw. D. Whether marked is a desirable outcome. 10 pts
8. The slides recommend plus-four when confidence is at least 90% and n>=10. Does n=24 at 85% confidence meet that recommendation?	A. Yes; n alone is sufficient. B. Yes; plus-four has no stated conditions. C. No; the confidence level is below the stated threshold. D. No; n must exceed 100. 10 pts
9. A recorded file size is far above the rest. What should happen before deleting it?	A. Delete it because all outliers are mistakes. B. Keep it only if the mean equals the median. C. Replace it with zero automatically. D. Check whether it is an error or a genuine unusual file. 10 pts
10. An exponential waiting time has mean 5 minutes. What rate per minute matches it?	A. 5. B. 0.5. C. 0.2. D. 25. 10 pts

65093

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

B.

10 pts

4. A.

10 pts

5.

B.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

C.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **77549**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Independent sample size increases. What happens to spread in original units?	A. SD of sums rises; SE of averages falls. B. Both rise. C. Both fall. D. Both stay unchanged. 10 pts
2. A is 6 above its mean with SD 3; B is 8 above its mean with SD 8. Which is more extreme above its group center?	A. The larger raw difference always determines extremeness. B. B. C. They are equally extreme. D. A. 10 pts
3. Which sequence gives the first success on trial 4?	A. Failure, failure, failure, success. B. Success, failure, failure, failure. C. Four successes. D. Failure, success, failure, success. 10 pts
4. Sample SD is 6 minutes and n=9; sqrt(9)=3. What estimated SE is used?	A. 18 minutes. B. 2 minutes. C. 3 minutes squared. D. 6 minutes. 10 pts
5. Which event is the complement of at least one failure in three trials?	A. Exactly one failure. B. At least two failures. C. All three failures. D. No failures. 10 pts
6. A bell-shaped distribution of scores has mean 50 and standard deviation 5. Approximately what percentage of scores lies between 45 and 55?	A. 68%. B. 50%. C. 95%. D. 99.7%. 10 pts
7. A cumulative table gives 30% below 5 seconds and 75% below 10 seconds. What percent is from 5 to less than 10?	A. 30%. B. 105%. C. 45%. D. 75%. 10 pts

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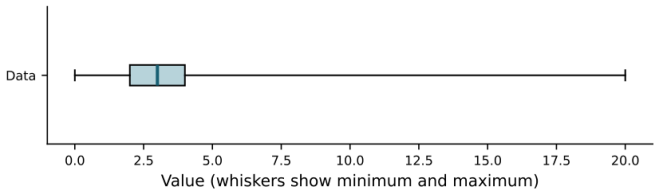
MSC 201 **77549**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A U(0,12) wait has already lasted 11 minutes. What is the largest possible additional wait under this model?	A. 11 minutes. B. 1 minute. C. There is no upper bound. D. 12 minutes. 10 pts
9. For a standard normal variable, compare P(Z<-1.4) and P(Z>1.4).	A. They are equal. B. Both equal 0.5. C. The right tail is larger. D. The left tail is larger. 10 pts
10. A basic box plot reports minimum 0, Q1=2, median 3, Q3=4, maximum 20. Which comparison is supported? 	A. The plot establishes the exact mean. B. The segment from Q3 to the maximum spans a wider numerical range than the segment from the minimum to Q1. C. The upper segment must contain three times as many observations. D. The lower segment contains no observations. 10 pts

77549

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. A.

10 pts

4. B.

10 pts

5. D.

10 pts

6. A.

10 pts

7. C.

10 pts

TEST PAGE 2

8. B.

10 pts

9. A.

10 pts

10. B.

10 pts

MSC 201 **10283**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. 40 of 100 sampled jobs succeed. What is the sample success proportion?	A. 0.60. B. 40. C. 2.5. D. 0.40. 10 pts
2. A question asks for the mass exceeded by only 10% of packages. What units should the answer have?	A. Mass units. B. No units because it is a z-score automatically. C. Probability units only. D. A percentage of sample size. 10 pts
3. A game charges a fixed entry fee and sometimes awards a prize. What distinguishes net gain from the prize amount?	A. The fee changes the chance of each prize. B. Net gain ignores outcomes with no prize. C. Net gain accounts for the fee paid to play. D. Net gain equals the largest possible prize. 10 pts
4. A well-run randomized experiment finds better results for an app among its volunteers. Which claim needs extra caution?	A. The observed volunteers had different average results. B. The same effect applies to all students everywhere. C. Random assignment helped compare the formats. D. The study provides evidence about an app effect in this setting. 10 pts
5. After an 11-minute wait, U(0,12) allows at most 1 more minute. An exponential model allows longer additional waits. What explains the difference?	A. Uniform waits are always zero. B. Both models are memoryless. C. Exponential waits stop at their mean. D. Uniform waits have a finite endpoint; exponential waits do not. 10 pts
6. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A one-sample t interval. B. A known-SD z interval treating s as known population SD. C. A proportion interval regardless of variable type. D. A prediction interval for a single future observation. 10 pts

Test continues on the other side of the sheet.

MSC 201 **10283**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A model allows counts 0,1,2,... without a finite upper limit. Can it still be discrete?	A. No; a count must stop at 10. B. Yes; countably many separated values can be discrete. C. Only if every count has equal probability. D. No; all infinite possibilities are continuous. 10 pts
8. Datasets share min=0 and max=100, but have different IQRs. What follows?	A. Their observations must be identical. B. Equal ranges do not imply equal middle-half spread. C. Their standard deviations must be identical. D. Their medians must be identical. 10 pts
9. A supplied sampling model gives P(mean<=12)=0.90. What is P(mean>12)?	A. 0.10. B. 0.90. C. 0.50. D. 1.90. 10 pts
10. The mean waiting count is 4 trials. Must the first success occur by trial 4?	A. Yes; the mean is a deadline. B. No; longer waits remain possible. C. No; it must occur exactly on trial 5. D. Yes; 4 is the maximum. 10 pts

10283

TEST PAGE 1

1.

D.

10 pts

2. A.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6. A.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

B.

10 pts

9. A.

10 pts

10.

B.

10 pts

MSC 201 **42305**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A first branch has probability 0.4; its next conditional branch is 0.5. What is the path probability?	A. 0.1. B. 0.9. C. 0.8. D. 0.2. 10 pts
2. X counts trials through first success. What does X>3 require?	A. A success on trial 3. B. Exactly three successes. C. Failures on the first three trials. D. At least one success in the first three trials. 10 pts
3. Ratings poor, fair, good, excellent are coded 1 through 4. What is safest to infer from the codes?	A. Excellent is exactly twice as favorable as fair. B. The categories have no order. C. Each adjacent rating differs by exactly one unit of satisfaction. D. Excellent ranks above good, but equal numerical gaps are not established. 10 pts
4. A fixed 5-kg container is added to a shipment of random item masses. What changes?	A. Both mean and SD rise by 5 kg. B. Expected total stays fixed. C. Total SD doubles. D. Expected total rises by 5 kg; total SD is unchanged. 10 pts
5. A dataset has mean 8. One observation below 8 is removed. What happens to the mean of the remaining values?	A. It becomes the removed value. B. It always stays 8. C. It increases. D. It decreases. 10 pts
6. For a normal model, where is the 10th percentile?	A. Above the mean. B. Below the mean. C. At the mean. D. At the largest possible value. 10 pts
7. A normal model has mean 50 and SD 2. Which interval contains about 99.7%?	A. 44 to 56. B. 48 to 52. C. 46 to 54. D. 50 to 56. 10 pts

Test continues on the other side of the sheet.

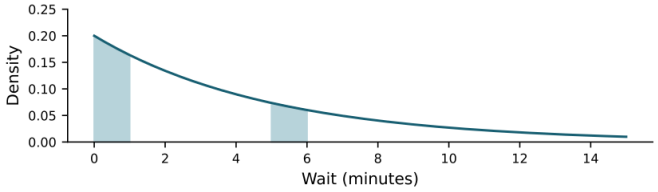
MSC 201 **42305**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. For an exponential wait, which one-minute interval has more probability? 	A. 0 to 1 minute. B. Both have probability zero. C. They have equal probability. D. 5 to 6 minutes. 10 pts
9. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 256. B. 15. C. 17. D. 16. 10 pts
10. A file is 60 KB; its group has mean 50 KB and SD 5 KB. How far above the mean is it?	A. 0.5 standard deviations. B. 2 standard deviations. C. 10 standard deviations. D. 12 standard deviations. 10 pts

42305

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6.

B.

10 pts

7. A.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

B.

10 pts

10.

B.

10 pts

MSC 201 **53724**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A box plot has median 6. Can it reveal exactly how many observations equal 5?	A. No; the summary does not retain those exact counts. B. Yes; exactly one quarter equal 5. C. Yes; none equal 5. D. Yes; half equal 5. 10 pts
2. The slides recommend plus-four when confidence is at least 90% and n>=10. Does n=24 at 85% confidence meet that recommendation?	A. Yes; plus-four has no stated conditions. B. No; the confidence level is below the stated threshold. C. Yes; n alone is sufficient. D. No; n must exceed 100. 10 pts
3. On a die, A is even and B is greater than 3. Which results satisfy both?	A. 2, 4, 5, 6. B. 5 only. C. 2 only. D. 4 and 6. 10 pts
4. The success chance improves after each failed attempt because the process learns. Why is a geometric model questionable?	A. The first success occurs eventually. B. There is a stopping rule. C. The success probability is not constant. D. Failures are counted. 10 pts
5. Three reported percentages are 33.3%, 33.3%, and 33.3%. What most reasonably explains the 99.9% total?	A. Rounding. B. Missing exactly one participant. C. The data must be fabricated. D. Overlapping categories are certain. 10 pts
6. To halve SE, how must an independent sample's size change?	A. Multiply n by 4. B. Add 4 regardless of its current size. C. Multiply n by 2. D. Divide n by 2. 10 pts
7. A nonnormal variable has a well-defined mean and SD. Can its z-score automatically be looked up as a standard-normal probability?	A. No; only normal variables can be standardized. B. Yes; a positive SD is sufficient. C. Yes; z-scores always imply normality. D. No; the normal shape assumption still matters. 10 pts

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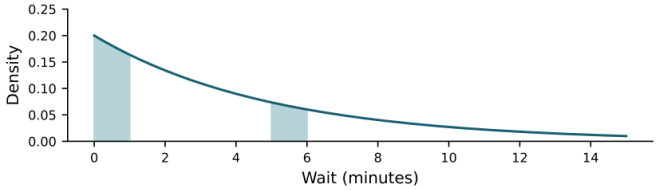
MSC 201 **53724**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Which t distribution has heavier tails?	A. df=4. B. They have equal tails. C. df=100. D. Both have no tails. 10 pts
9. Four cables are sampled uniformly without replacement from 12, of which 5 are faulty. Which model directly fits the faulty count?	A. Normal individual measurement. B. Continuous uniform. C. Hypergeometric. D. Geometric waiting count. 10 pts
10. For an exponential wait, which one-minute interval has more probability? 	A. 5 to 6 minutes. B. 0 to 1 minute. C. Both have probability zero. D. They have equal probability. 10 pts

53724

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. D.

10 pts

4. C.

10 pts

5. A.

10 pts

6. A.

10 pts

7. D.

10 pts

TEST PAGE 2

8. A.

10 pts

9. C.

10 pts

10. B.

10 pts

MSC 201 **60667**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Which shortcut finds at least one success in a fixed set of independent trials?	A. The probability of exactly one success only. B. 1 minus the probability all succeed. C. 1 minus the probability of no successes. D. The expected success count. 10 pts
2. A wait is uniform from 0 to 12 minutes. What is P(3<X<6)?	A. 9/12. B. 3/12. C. 6/12. D. 3/6. 10 pts
3. A model describes the mass of one package. Is its SD automatically appropriate for the mean mass of 100 independent packages?	A. Yes; averages always vary more. B. Yes; both are measured in grams. C. No; the average has less spread. D. No; averages have no units. 10 pts
4. Notebook colors are blue, blue, black, red. Which statement meaningfully summarizes a typical category?	A. The mean color is 2. B. Blue is the most common color. C. The standard deviation of color names is 1. D. The median is 2.5 colors. 10 pts
5. Independent trials have a constant chance of success strictly between zero and one. Which variable can take arbitrarily large values?	A. The number of trials through the first success. B. The number of successes in a fixed set of trials. C. The number of outcomes on one yes-or-no trial. D. The number of marked items in a fixed-size sample. 10 pts
6. A fair coin gives 8 heads in 10 tosses. What follows?	A. The next two tosses must be tails. B. Such a short-run imbalance can occur with a fair coin. C. The coin's future head probability is now 0.8. D. Fairness requires exactly 5 heads in every ten tosses. 10 pts

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7. A mean interval is 10 to 14 minutes. Which report matches its target?	A. Every future duration will be between 10 and 14. B. The procedure estimates the population mean duration between 10 and 14 minutes. C. The median is guaranteed to equal 12. D. Exactly 95 of the next 100 durations will lie there. 10 pts
8. Frequencies for 0, 1, 2, and 3 jobs are 2, 3, 4, and 1. How many sessions had at most 1 job?	A. 2. B. 3. C. 5. D. 9. 10 pts
9. Twenty items have sample mean mass 5 g. What is their total mass?	A. 25 g. B. 100 g. C. 0.25 g. D. 5 g. 10 pts
10. 400 people are sampled without replacement from 20,000. Does this meet the at-most-10% sample-fraction guideline?	A. No; n must be below 30. B. Yes; any random sample meets it. C. No; the fraction is 20%. D. Yes; the fraction is 2%. 10 pts

60667

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5. A.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

C.

10 pts

9.

B.

10 pts

10.

D.

10 pts

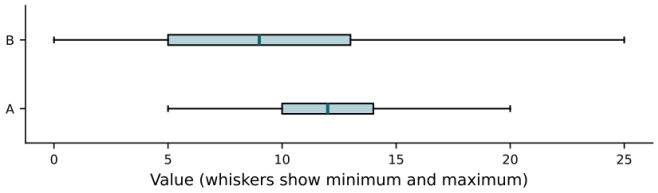
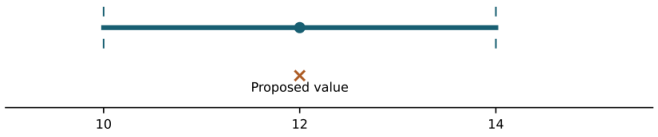
MSC 201 **75628**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Why represent a single binomial count by an interval when using a normal approximation?	A. The count can take fractional values. B. An interval has area under a continuous curve, while a single point has zero area. C. A count always equals its expected value. D. The normal distribution has positive probability at every exact point. 10 pts
2. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A prediction interval for a single future observation. B. A proportion interval regardless of variable type. C. A one-sample t interval. D. A known-SD z interval treating s as known population SD. 10 pts
3. Box plot A has median 12, IQR 4; B has median 9, IQR 8. Which statement is supported? 	A. A has higher median and smaller middle-half spread. B. B has higher median and smaller spread. C. A has lower median and larger spread. D. Every value in A exceeds every value in B. 10 pts
4. A bar graph of counts 98 and 100 starts its vertical axis at 97. What impression might it exaggerate?	A. The size of the difference between the counts. B. The fact that 98 is below 100. C. The number of categories. D. The total number of observations. 10 pts
5. A mean interval is 10 to 14. How should a proposed mean of 12 be described? 	A. Compatible with the interval, without being proved exact. B. Proved to be the exact population mean. C. The only compatible value. D. Impossible. 10 pts

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MSC 201 **75628**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

6. A proposed normal model gives substantial probability to negative durations. What is the concern?	A. Its mean must be zero. B. Normal distributions cannot model any positive variable. C. Every duration must be negative. D. It assigns appreciable probability to impossible values. 10 pts
7. Plan A checks 10 items and counts successes. Plan B checks until the first success and counts trials. Which is geometric under independent constant-p trials?	A. Plan B. B. Both must be geometric. C. Neither because trials are discrete. D. Plan A. 10 pts
8. A buyer counts rejected items among a fixed number of independent inspections with the same rejection chance. What should be called a success for this model?	A. A rejection, because that is what the variable counts. B. An acceptance, because success must be favorable. C. A change in the rejection chance. D. A set with no rejections. 10 pts
9. A service meets its deadline with probability 0.80. What is the probability of missing it?	A. 1.80. B. 0.40. C. 0.80. D. 0.20. 10 pts
10. Uniform model A spans width 4; model B spans width 8. Which density is taller?	A. B. B. The one with the larger mean regardless of width. C. They have equal height. D. A. 10 pts

75628

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3. A.

10 pts

4. A.

10 pts

5. A.

10 pts

TEST PAGE 2

6.

D.

10 pts

7. A.

10 pts

8. A.

10 pts

9.

D.

10 pts

10.

D.

10 pts

MSC 201 **54550**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

1. A uniform interval centered at 10 doubles its width while keeping center 10. What happens?	A. Mean stays 10; SD doubles. B. Mean doubles; SD stays unchanged. C. Both stay unchanged. D. Both double. 10 pts
2. A known-SD mean interval currently has margin 1. To target margin 0.5 at the same confidence, how should n change?	A. Add 1 to n. B. Halve n. C. Multiply n by 4. D. Multiply n by 2. 10 pts
3. A normal model's SD doubles with mean fixed. What happens to its model IQR?	A. It quadruples. B. It doubles. C. It becomes zero. D. It stays unchanged. 10 pts
4. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A known-SD z interval treating s as known population SD. B. A prediction interval for a single future observation. C. A proportion interval regardless of variable type. D. A one-sample t interval. 10 pts
5. A model has one blank probability entry and lists every possible value. What does the blank need to account for?	A. Half the total probability, regardless of the other entries. B. The average of the listed probabilities. C. All probability not already assigned to the other values. D. The probability of the most likely listed value. 10 pts
6. A mean is estimated from grouped intervals using midpoints. Why is it generally approximate?	A. The sample size is necessarily unknown. B. Midpoints are never numerical values. C. Frequencies cannot be used in means. D. Actual observations need not equal the interval midpoints. 10 pts

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MSC 201 **54550**

Final Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

7. A small prize is common and a large prize is rare. Why is the midpoint between the prizes generally a poor estimate of the expected payout?	A. The small prize does not count toward the average. B. The expected payout must equal the larger prize. C. It gives the two prizes equal weight despite their different chances. D. Only the larger prize determines the mean. 10 pts
8. Someone says n=30 guarantees normal sample means for every population. What is the problem?	A. Thirty is too large for any sample. B. Only populations with mean zero qualify. C. Sample means can never approach normality. D. Skewness and extreme observations affect how large n needs to be. 10 pts
9. A grouped table reports 12 durations from 5 to less than 10 seconds. Can it tell exactly how many equal 7 seconds?	A. Yes; none may equal 7. B. Yes; all 12 must equal 7. C. Yes; exactly half must equal 7. D. No; exact positions within the interval are hidden. 10 pts
10. Which sequence gives the first success on trial 4?	A. Success, failure, failure, failure. B. Failure, success, failure, success. C. Four successes. D. Failure, failure, failure, success. 10 pts

54550

TEST PAGE 1

1. A.

10 pts

2. C.

10 pts

3. B.

10 pts

4. D.

10 pts

5. C.

10 pts

6. D.

10 pts

TEST PAGE 2

7. C.

10 pts

8. D.

10 pts

9. D.

10 pts

10. D.

10 pts